

Charles Phu

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EXPERIENCE

Headstarter AI

July 2024 – September 2024

Software Engineering Fellow

- Developed and deployed 5 AI projects in 5 weeks using **React JS**, **Next.js**, **Firebase**, and **Vercel**
- Collaborated in a team of 4 to develop an interactive customer support agent using Next.js, integrated a custom RAG pipeline using **OpenAI** and **Pinecone** that responds based on a company's knowledge base
- Engineered a SaaS product that generates dynamic flashcards based on any topic using the **Llama 3.1 LLM**, integrated a paywall and custom pricing plans using the **Stripe API**
- Implemented a web scraper to extract data from Rate my Professor and upsert to a **Pinecone** index, integrated with a RAG pipeline using **LangChain** and **OpenAI GPT-4o** to get up-to-date and relevant answers to user queries

Outlier AI

June 2024 – Present

Machine Learning Validation Engineer

- Evaluated and compared multiple LLM outputs to determine task suitability, improving LLM selection efficiency
- Validated model outputs for accuracy, reliability, and grammatical correctness, ensuring high-quality AI responses

EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

Bachelor of Science in Computer Science, **GPA: 3.84**

Expected June 2025

Relevant Coursework: Applied Machine Learning: Deep Learning, Artificial Intelligence, Data Structures and Algorithms, Principles of Computer Systems Design, Computational Models, Database Systems

SKILLS

Programming Languages: Python, C/C++, Lua/Luau, HTML/CSS, JavaScript, SQL

Frameworks/Libraries: React.js, Next.js, Material-UI, Scikit-learn, Regex, Pandas, Pytorch, PostgreSQL

PROJECTS

Roblox 2v2 team space game | *ROBLOX studio, Lua, Scripting, Modeling* | roblox.com/games/8715055706/Solar-Strife

- Team-based strategic PVP sci-fi game featuring fully armed spacecraft with gyroscopic movement
- Custom-modeled planes, ships, bases, and turrets with fully functional ship and player combat system

SpeedFill | *Javascript, Next.js, Supabase, Vercel, Gemini, Figma* | <https://speed-fill.vercel.app/>

- Product Owner and Manager**
- Developed a website to analyze how well resumes match up with job descriptions, generate cover letters, and receive AI-feedback
- Led the implementation of authentication, data storage, and UI components while optimizing backend processing and user experience

AI Form Detector | *YoloV8, Transfer Learning, Computer Vision*

- Created an AI model to validate squat, bench, and deadlift form by using YOLOv8 achieving 85% accuracy in classification

Multithreaded HTTP Server | *C programming, Regex*

- Designed and implemented a high-performance multithreaded HTTP server, enabling concurrent client request handling with robust multithreaded thread safety
- Developed a bounded buffer using reader/writer locks and semaphores to ensure efficient and thread-safe management of client requests

Hackathon Healthbot Website | *HTML, CSS, Javascript*

- 2nd place winner in Equal Opportunity StarHacks Online Hackathon
- Designed and created a website to entertain people during COVID

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Upper Division Tutor: Principles of Computer System Design

September 2024 – Present

University of California, Santa Cruz

- Provided one-on-one support to students, resolving programming issues and debugging code to improve assignment outcomes
- Assisted students with problem-solving and code debugging in programming assignments, ensuring efficient and effective solutions